

Exercise 1: Configuring a Basic Spring Application

pom.xml

```
<dependencies>
  <dependency>
    <groupId>org.springframework</groupId>
    <artifactId>spring-context</artifactId>
    <version>5.3.30</version>
  </dependency>
</dependencies>
```

applicationContext.xml

```
<beans xmlns="http://www.springframework.org/schema/beans">

  <bean id="bookRepository"
        class="com.library.repository.BookRepository"/>

  <bean id="bookService"
        class="com.library.service.BookService"/>

</beans>
```

BookRepository.java

```
package com.library.repository;

public class BookRepository {

    public void display() {
        System.out.println("Book Repository");
    }
}
```

BookService.java

```
package com.library.service;

public class BookService {

    public void service() {
        System.out.println("Book Service");
    }
}
```

Main Class

```
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class LibraryManagementApplication {

    public static void main(String[] args) {

        ApplicationContext context =
            new ClassPathXmlApplicationContext(
                "applicationContext.xml");

        BookService service =
            context.getBean("bookService",
                BookService.class);

        service.service();
    }
}
```

Exercise 2: Dependency Injection

BookService.java

```
package com.library.service;

import com.library.repository.BookRepository;

public class BookService {

    private BookRepository repository;

    public void setRepository(
        BookRepository repository) {

        this.repository = repository;
    }

    public void service() {
        repository.display();
    }
}
```

applicationContext.xml

```
<bean id="bookRepository"
      class="com.library.repository.BookRepository"/>

<bean id="bookService"
      class="com.library.service.BookService">

    <property name="repository"
        ref="bookRepository"/>
</bean>
```

Exercise 3: Logging with Spring AOP

pom.xml

```
<dependency>
    <groupId>org.springframework</groupId>
    <artifactId>spring-aop</artifactId>
    <version>5.3.30</version>
</dependency>

<dependency>
    <groupId>org.aspectj</groupId>
    <artifactId>aspectjweaver</artifactId>
    <version>1.9.20</version>
</dependency>
```

LoggingAspect.java

```
package com.library.aspect;

import org.aspectj.lang.ProceedingJoinPoint;

public class LoggingAspect {

    public Object logExecutionTime(
        ProceedingJoinPoint joinPoint)
        throws Throwable {

        long start = System.currentTimeMillis();

        Object result = joinPoint.proceed();

        long end = System.currentTimeMillis();

        System.out.println(
```

```
        "Execution Time : " +
        (end - start) + " ms");

    return result;
}
}
```

Exercise 4: Creating Maven Project

pom.xml

```
<dependencies>

    <dependency>
        <groupId>org.springframework</groupId>
        <artifactId>spring-context</artifactId>
        <version>5.3.30</version>
    </dependency>

    <dependency>
        <groupId>org.springframework</groupId>
        <artifactId>spring-aop</artifactId>
        <version>5.3.30</version>
    </dependency>

    <dependency>
        <groupId>org.springframework</groupId>
        <artifactId>spring-webmvc</artifactId>
        <version>5.3.30</version>
    </dependency>

</dependencies>

<build>
<plugins>
    <plugin>
        <groupId>
            org.apache.maven.plugins
        </groupId>

        <artifactId>
            maven-compiler-plugin
        </artifactId>

        <configuration>
            <source>1.8</source>
```

```
        <target>1.8</target>
    </configuration>
</plugin>
</plugins>
</build>
```

Exercise 5: Spring IoC Container

applicationContext.xml

```
<bean id="bookRepository"
      class="com.library.repository.BookRepository"/>

<bean id="bookService"
      class="com.library.service.BookService">

    <property name="repository"
              ref="bookRepository"/>
</bean>
```

BookService.java

```
public class BookService {

    private BookRepository repository;

    public void setRepository(
        BookRepository repository) {

        this.repository = repository;
    }
}
```

Exercise 6: Annotation-Based Configuration

applicationContext.xml

```
<context:component-scan
base-package="com.library"/>
```

BookRepository.java

```
@Repository
public class BookRepository {
}
```

BookService.java

```
@Service
public class BookService {
}
```

Exercise 7: Constructor and Setter Injection

BookService.java

```
public class BookService {

    private BookRepository repository;

    public BookService(
        BookRepository repository) {

        this.repository = repository;
    }

    public void setRepository(
        BookRepository repository) {

        this.repository = repository;
    }
}
```

applicationContext.xml

```
<bean id="bookRepository"
    class="com.library.repository.BookRepository"/>

<bean id="bookService"
    class="com.library.service.BookService">

    <constructor-arg
        ref="bookRepository"/>

    <property name="repository"
        ref="bookRepository"/>
</bean>
```

Exercise 8: Basic AOP with Spring

LoggingAspect.java

```
@Aspect
public class LoggingAspect {

    @Before(
        "execution(* com.library.service.*.*(..))")
    public void beforeMethod() {

        System.out.println(
            "Before Method Execution");
    }

    @After(
        "execution(* com.library.service.*.*(..))")
    public void afterMethod() {

        System.out.println(
            "After Method Execution");
    }
}
```

applicationContext.xml

```
<aop:aspectj-autoproxy/>

<bean id="loggingAspect"
    class="com.library.aspect.LoggingAspect"/>
```

Exercise 9: Spring Boot Application

application.properties

```
spring.datasource.url=jdbc:h2:mem:testdb
spring.datasource.driverClassName=org.h2.Driver

spring.jpa.hibernate.ddl-auto=update
```

Book Entity

```
@Entity
public class Book {

    @Id
    private Long id;
```

```
    private String title;
    private String author;
}
```

Repository

```
@Repository
public interface BookRepository
extends JpaRepository<Book, Long> {
}
```

Controller

```
@RestController
@RequestMapping("/books")
public class BookController {

    @Autowired
    private BookRepository repository;

    @GetMapping
    public List<Book> getAllBooks() {
        return repository.findAll();
    }

    @PostMapping
    public Book addBook(
        @RequestBody Book book) {

        return repository.save(book);
    }
}
```

Main Class

```
@SpringBootApplication
public class LibraryManagementApplication {

    public static void main(
        String[] args) {

        SpringApplication.run(
            LibraryManagementApplication.class,
            args);
    }
}
```